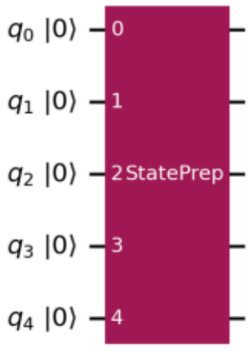
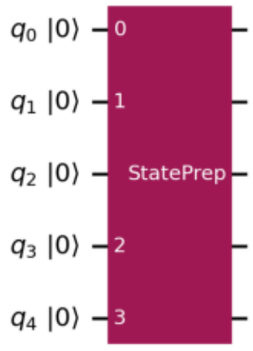
Dense vs structured sparse amplitude encoding — Qiskit circuits

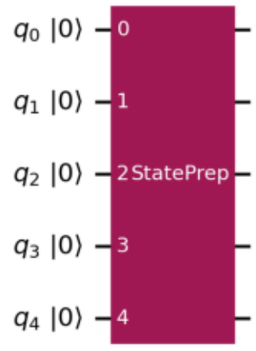
ToN-IoT | Before: dense (32 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 100



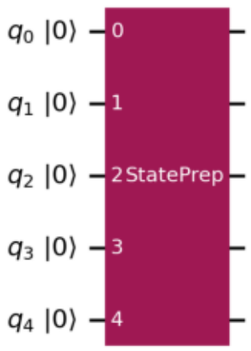
ToN-IoT | After: sparse (at most 16 amplitudes)
Active prep qubits: 4/5 | CX: 11 | depth: 35



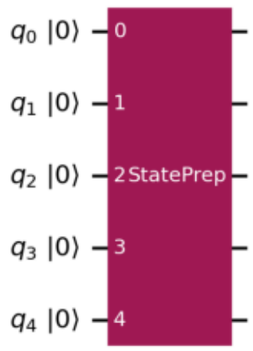
ToN-IoT | After: sparse (at most 8 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 102



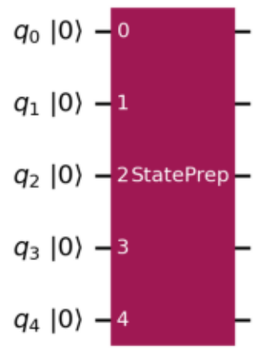
RT-IoT | Before: dense (32 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 95



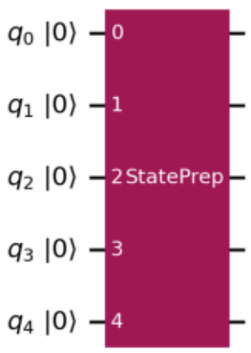
RT-IoT | After: sparse (at most 16 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 102



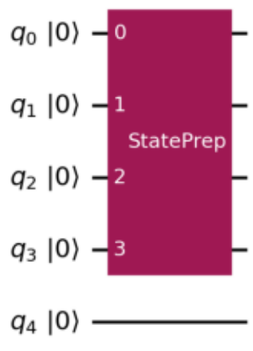
RT-IoT | After: sparse (at most 8 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 89



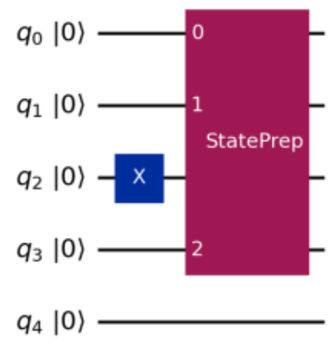
NSL-KDD | Before: dense (32 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 101



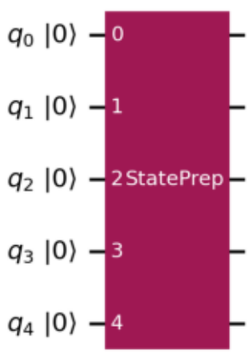
NSL-KDD | After: sparse (at most 16 amplitudes)
Active prep qubits: 4/5 | CX: 11 | depth: 35



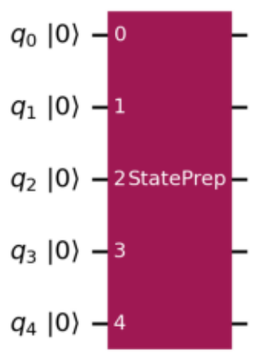
NSL-KDD | After: sparse (at most 8 amplitudes)
Active prep qubits: 3/5 | CX: 4 | depth: 14



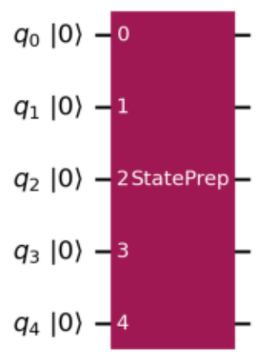
UNSW-NB15 | Before: dense (32 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 99



UNSW-NB15 | After: sparse (at most 16 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 99



UNSW-NB15 | After: sparse (at most 8 amplitudes)
Active prep qubits: 5/5 | CX: 26 | depth: 107



Same first training sample within each dataset; all circuits retain 5 wires. StatePrep blocks are expanded in the accompanying gate-level files.
CX/depth refer to logical state preparation in basis {rx, ry, rz, h, cx}; no hardware routing, noise channels, or measurement gates.